Faculty of Health and Medical Sciences

Day 2: Baseline table, covariate adjustment with binary outcome & robustness

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Outline/Intended Learning Outcomes (ILOs)

Baseline table

- ILO: repeat the main recommendations
- ILO: explain their rationale to surprised colleagues and reviewers
- ILO: describe the relationship between randomization, baseline imbalance and the validity of statistical analyses

Covariate adjustment

- ILO: repeat the main recommendations
- ILO: explain how this can lead to power gains
- ILO: restate key robustness properties
- ILO: perform conventional analyses for binary outcomes
- ILO: recall examples

Baseline Tables

- Baseline tables are usually presented in articles presenting the results of a clinical trial.
- Guidelines (e.g., CONSORT) encourage us to present them.[1]

CONSORT 2010 checklist of information to include when reporting a randomised trial*

Section/Topic	Item No	Checklist item	Reported on page No
Title and abstract			
	1a	Identification as a randomised trial in the title	
	1b	Structured summary of trial design, methods, results, and conclusions (for specific guidance see CONSORT for abstracts [45,65])	
Introduction			
Background and objectives	2a	Scientific background and explanation of rationale	
	2b	Specific objectives or hypotheses	
Methods			
Trial design	3a	Description of trial design (such as parallel, factorial) including allocation ratio	
Results			
Participant flow (a diagram is strongly recommended)	13a	For each group, the numbers of participants who were randomly assigned, received intended treatment, and were analysed for the primary outcome	
	13b	For each group, losses and exclusions after randomisation, together with reasons	
Recruitment	14a	Dates defining the periods of recruitment and follow-up	
	14b	Why the trial ended or was stopped	
Baseline data	15	A table showing baseline demographic and clinical characteristics for each group	

[1] Moher et al (2010). CONSORT 2010 Explanation and Elaboration: updated guidelines for reporting parallel group randomised trials. Journal of Clinical Epidemiology, 63(8), e1–e37

Why presenting a baseline table?

> *5.4.4. Item 15. A table showing baseline demographic and clinical characteristics for each group*
>
> Example—See Table 4
>
> Explanation—Although the eligibility criteria (see item 4a) indicate who was eligible for the trial, it is also important to know the characteristics of the participants who were actually included. This information allows readers, especially clinicians, to judge how relevant the results of a trial might be to an individual patient.

Why presenting a baseline table?

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What to put in a baseline table? Simple descriptive statistics (only!).

> ported, along with average values. Continuous variables can be summarised for each group by the mean and standard deviation. When continuous data have an asymmetrical distribution, a preferable approach may be to quote the median and a centile range (such as the 25th and 75th centiles) [177]. Standard errors and confidence intervals are not appropriate for describing variability—they are inferential rather than descriptive statistics. Variables with

Source: Moher et al (2010). CONSORT 2010 Explanation and Elaboration: updated guidelines for reporting parallel group randomised trials. Journal of Clinical Epidemiology, 63(8), e1–e37.

Example

	Initial oral antibiotics (n=101)	Initial intravenous antibiotics (n=91)
Sex		
Female	37 (37%)	31 (34%)
Male	64 (63%)	60 (66%)
Age, years	2·4 (1·3-6·7)	2·3 (1·4-6·7)
0–4	66 (65%)	63 (69%)
5–9	20 (20%)	13 (14%)
10–17	15 (15%)	15 (17%)
Localisation		
Bone infection	41 (41%)	41 (45%)
Joint infection	43 (43%)	36 (40%)
Spondylodiscitis or sacroiliitis	17 (17%)	14 (15%)
Clinical presentation		
Symptom duration before treatment, days	5·0 (3·0-12·0)	5·0 (2·5-10·5)
Pain or immobilisation	101 (100%)	90 (99%)
Temperature ≥38·0 °C	66 (65%)	60 (66%)
Local signs of infection	41 (41%)	42 (46%)
C-reactive protein, mg/L		
At study enrolment	34 (10-65)	34 (14-58)
Diagnostic evaluations		
Blood culture	94 (93%)	88 (97%)
Joint puncture	35 (35%)	25 (27%)
Bone biopsy	4 (4%)	4 (4%)
MRI	77 (76%)	63 (69%)
Ultrasound	77 (76%)	72 (79%)
X-ray	83 (82%)	69 (76%)
Pathogen identified (sterile site)		
All pathogens†	25 (25%)	23 (25%)
S aureus‡	16 (16%)	13 (14%)
K kingae§	5 (5%)	8 (9%)
S pyogenes	4 (4%)	1 (1%)
S pneumoniae	0	1 (1%)

"Data are n (%) or median (IQR)."[2]

[2] Nielsen et al. (2024). Oral versus intravenous empirical antibiotics in children and adolescents with uncomplicated bone and joint infections: a nationwide, randomised, controlled, non-inferiority trial in Denmark. Lancet Child Adolesc Health, 4642(24), 1–11.

Don't add p-values to baseline tables!

- Although it is commonly done and also commonly asked by reviewers, unfortunately. But, usually not by statistical reviewers!
- CONSORT guidelines do not support adding such p-values:

  "Such hypothesis testing is superfluous and can mislead investigators and their readers." [3]

  (most medical journals encourage authors to follow the CONSORT guidelines; check their website)
- In the website of the *New England Journal of Medicine*, section "Statistical Reporting Guidelines", it is even stated:[4]

> b. P values should not be included in the traditional Table 1 of a randomized trial manuscript showing the distribution of baseline variables by treatment group. However, authors should note imbalances in potential confounders that could be due to chance or inconsistencies in randomization.

[3] Moher et al (2010). CONSORT 2010 Explanation and Elaboration: updated guidelines for reporting parallel group randomised trials. Journal of Clinical Epidemiology, 63(8), e1–e37

[4] accesssed Oct 29, 2025: https://www.nejm.org/author-center/new-manuscripts#statistical-reporting-guidelines

Why is it irrational to add p-values to baseline tables?

"Hardly a statistician of repute can be found to defend the practice common among physicians of comparing the treatment groups in a randomized clinical trial at baseline using hypothesis/significance tests on covariates. The reason for the statistician's dislike is that such a test appears to be used to say something about the adequacy of the given allocation whereas it could only be a test of the allocation procedure: the randomization process itself."

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Can't I do like many others and add them anyway?

"In short, the test of baseline balance is a misuse of the significance test. The fact that it is frequently performed does not constitute a defence any more than the fact that antibiotics are commonly employed to 'treat' viral infections proves that they are effective antivirals. And the fact that baseline tests are commonly performed without much apparent harm is no more of a defence than saying of the policy of treating colds with antibiotics that most patients recover."

Ref: quotes from Section 7.2.1, pages 112-113, in *Statistical Issues in Drug Development* (3rd Edition), by Stephen Senn.



Does random imbalance invalidate the results?

This seems to be a common misunderstanding!

"It is not necessary for the groups to be balanced. In fact, the probability calculation applied to a clinical trial automatically makes an allowance for the fact that groups will almost certainly be unbalanced, and if one knew that they were balanced, then the calculation that is usually performed would not be correct." [5]

In other words, standard errors, confidence intervals and p-values are computed accounting for the fact that we expect some random imbalance.



[5] Senn S. Seven myths of randomisation in clinical trials. Stat Med 2013;32:1439e50.

Random imbalance and trial sample size

Question: Some think that larger trials are generally less vulnerable to covariate imbalance than smaller ones. Is that really true?



[6] Senn S. Seven myths of randomisation in clinical trials. Stat Med 2013;32:1439e50.

Random imbalance and trial sample size

Question: Some think that larger trials are generally less vulnerable to covariate imbalance than smaller ones. Is that really true?

Answer: No, not if look at confidence intervals and p-values, i.e., if we look at results that rigorously account for randomness /uncertainty.

"When sample sizes increase, it is certainly the case that the expected random difference between two groups will reduce, and this reflects, amongst other things, the greater expected balance in proportionate terms between groups. In this sense, the belief that larger trials are more balanced than smaller ones is not a myth. However, by the same token, the standard error of the treatment effect will be smaller and the confidence interval will be narrower, and for any given observed difference at outcome, the p-value will be smaller. Thus, the effect of increasing sample size is consumed by conventional analyses in terms of increased precision."[6]



[6] Senn S. Seven myths of randomisation in clinical trials. Stat Med 2013;32:1439e50.

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Rationale for baseline covariate adjustment

Should baseline covariates be ignored because of randomization?

"one should not use the fact that one has randomized as an excuse for ignoring baseline prognostic information in analyzing a clinical trial. Such prognostic information provides the means of distinguishing the particular clinical trial from the much larger set of trials one might have run in which the baseline prognostic distribution might have been different." [7]

Baseline covariates provides important information that can lead to power gains, when this information is used well in the statistical analysis, using so-called "adjusted analyses".

"incorporating prognostic baseline covariates in the design and analysis of clinical trial data can result in a more efficient use of data to demonstrate and quantify the effects of treatment"[8]



[7] Senn. (2005). Baseline balance and valid statistical analyses: common misunderstandings. Applied Clinical Trials 14 (3): 24–27.

[8] FDA scientific guidelines: "Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products Guidance for Industry", 2023, accessed 29/10/2025, https://www.fda.gov/media/148910/download

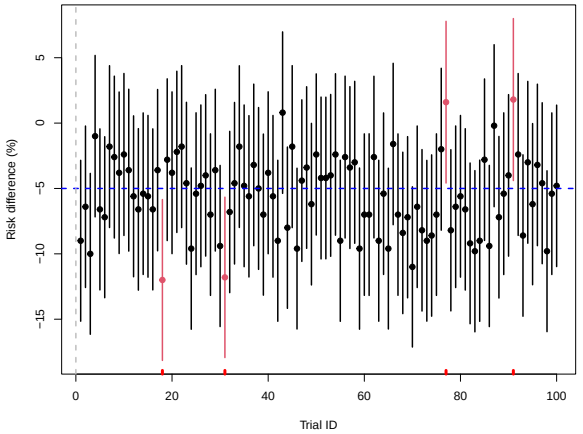
Why are adjusted analyses more powerful? (1/4)

- Randomization balances covariates in average
- But in average only! For any trial, there will be some degree of imbalance.
- If some covariates are prognostic of the outcome, random baseline covariates imbalance creates random variation, when we estimate a treatment effect, e.g., a risk difference with binary outcomes.
- When we adjust for baseline covariates, we (implicitely) compare subjects between arms within subgroups of subjects who have the same covariates profile. Within these subgroups, covariates balance necessarily holds. This is similar to the situation where we stratify the randomization.
- When covariates balance is enforced, there is less random variation, which results in smaller SEs (when computed appropriately) and more power.

Conclusion: when performing an unadjusted analysis, we make an allowance for the fact that there will be random covariates imbalance. But when we adjust we don't, as we do not need, and that is how we gain power.



Why are adjusted analyses more powerful? (2/4)



Setting:
- 100 trials simulated
- shown are 95%-CI; unadjusted analysis
- $\approx$ 95 out 100 contain the "true" risk difference (blue line), as expected
- Those who do not are shown in red
- mean(SE)=3.15%

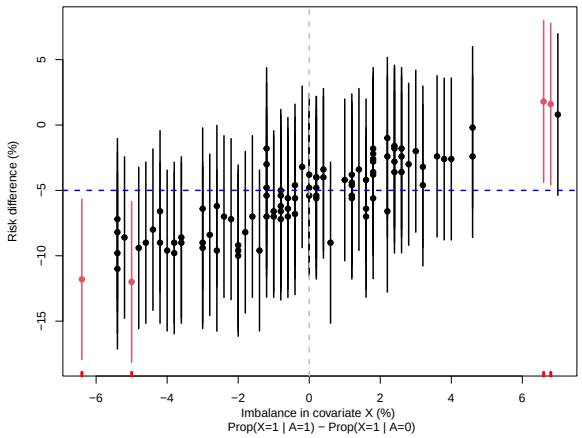
Details:
- $n = 500$ per group, randomized 1:1
- One binary covariate X ($p_X = 0.5$)
- Marginal (average) risks: $\pi_1 = 50\%$ vs $\pi_0 = 55\%$, from 4 subgroups, 25% prevalence for each.

A (intervention)	X (severity)	risk of death (90-day)
0	0	11%
1	0	10%
0	1	99%
1	1	90%

- $RR_{X|A} = 9$ and $RR_{A|X} = 0.91$



Why are adjusted analyses more powerful? (3/4)

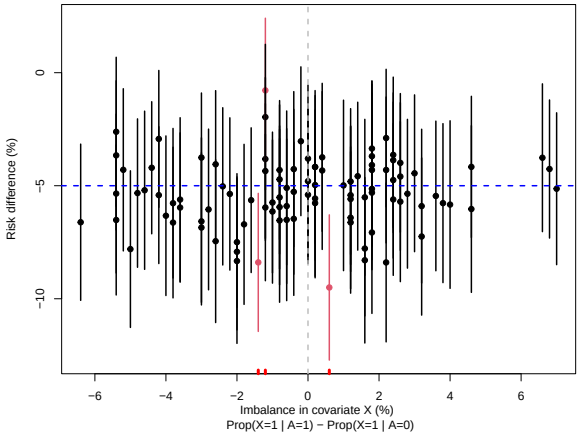


- shown are 95%-CI; unadjusted analysis
- imbalance ↑ ⇒ estimated risk difference ↑
- it reflects that X is prognostic of the outcome ($X \uparrow \Rightarrow$ risk ↑)
- red CIs that do not include the truth corresponds to large imbalance.
- Random large imbalance corresponds to a situation randomly far from the average, as in average there is balance.
- The CIs are on average centered around the truth
- But there are NOT centered in average, within the subgroup of "atypical" trials with large positive (or large negative) unbalance.

"To use some statistical jargon, the baselines provide a means of identifying a recognizable subset, a group for which the average no longer applies. An analogy from life insurance may be helpful here. A life-table may provide a reasonable assessment of the expectation of life of a male aged 50. However, if it is based on an "average" population, then it will overstate the expectation, other things being equal, for a 50-year-old male who is a smoker and underestimate it if he is not." [9]

[9] Senn (2019). The well-adjusted statistician: Analysis of covariance explained, Applied Clinical Trials.

Why are adjusted analyses more powerful? (4/4)



- shown are 95%-CI; adjusted analysis (logistic regression followed by standardization; non-parametric Bootstrap to compute SEs, n.boot=400)
- No association between imbalance and estimated risk difference
- 97/100 contain the "true" risk difference (blue line), approximately as the 95 expected (random, because only 100 simulated datasets)
- Those who do not are shown in red; random location.
- Note: mean(SE)=1.70% instead of mean(SE)=3.15% when unadjusted analysis. Much narrower CIs and more power !
- Corresponds to a redution in sample size needed of ≈ 70%, in this *extreme* example ($RR_{X|A} = 9$); because 1- $(\text{Var}_a/\text{Var}_u) \approx 0.70$.

How to adjust? (binary outcome)

- Different approach exist to analyze the data while adjusting for covariates.
- One which is increasingly promoted (and which was used in the previous example with the simulations) is "Regression standardization". This approach is often used in epidemiology to account for counfounding[10]. This approach is also termed the 'standardized', 'plug-in', or 'g- computation' estimator.[11]
- This approach targets an unconditional (also called marginal, or average) treatment effect, often quantified as a risk difference. That is, it targets the same as an unadjusted analysis which simply compare the proportions in the two arms.
- By contrast, usual multiple (aka adjusted) logistic regression targets a conditional treatment effect, quantified via odds ratios. That is, the same as odds ratio computed within strata of baseline variables.

[10] Sjölander. "Regression standardization with the R package stdReg." European Journal of Epidemiology 31.6 (2016): 563-574.

[11] FDA scientific guidelines: "Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products Guidance for Industry", 2023, accessed 29/10/2025, https://www.fda.gov/media/148910/download

Conditional vs unconditional effect

"In general, treatment effects may differ across subgroups. However, with some parameters such as odds ratios, even when all subgroup treatment effects are identical, this subgroup-specific conditional treatment effect can differ from the unconditional treatment effect (i.e., the effect at the population level from moving the target population from untreated to treated) (Gail et al. 1984). This is termed non-collapsibility (Agresti 2002), which is distinct from confounding and can occur despite randomization and large sample sizes" [12]

Table 1: Non-collapsibility of the Odds Ratio in a Hypothetical Target Population

	Percentage of target population	Success rate		Odds ratio
		New drug	Placebo	
Biomarker-positive	50%	80.0%	33.3%	8.0
Biomarker-negative	50%	25.0%	4.0%	8.0
Combined	100%	52.5%	18.7%	4.8

Reminder: odds = risk/(1-risk), Odds ratio= ratio of odds (in the two arms). E.g, $\frac{0.8}{0.2} / \frac{0.333}{0.667} = 8.0$.

[12] FDA scientific guidelines: "Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products Guidance for Industry", 2023, accessed 29/10/2025, https://www.fda.gov/media/148910/download

Digression: how usfeul are odds ratio?

- The relative merits of odds ratios compared to those of risk ratios and risk differences (and number needed to treat, NNT) have long been debated.[13]
- Many have pointed out that odds ratios are difficult to understand (and often misunderstood) and that reporting odds ratios should be avoided.[14]
- Others have replied that they are nevertheless very useful

> *"The principal benefit of ORs is that the assumption of homogeneity of ORs is a more tenable assumption than the assumption of homogeneity of RRs and thus it is more likely that an estimate of the OR can be reliably applied to all individuals within a population"*[15]

- And also that challenges to interpret them is not a good argument to not use them.

> *"Another argument that is sometimes advanced in favour of NNTs is that they are simple to interpret. The counterargument would be that a simple lie is being preferred to a complex truth. As Hutton has pointed out, differential equations are a difficult topic to under- stand but nobody suggests that their use should be removed from pharmacokinetics (Hutton, J.L. 2000)".*[16]

- Some also pointed out that risk ratios seem intuitive but are actually (often) misleading.

> *"Risk ratios may seem intuitive and easily applied; however, they are easily misapplied and the conclusions drawn from their use may be inappropriate. An intuitive, easily understood summary measure is worthwhile only to the extent that it results in valid conclusions."* [read the paper for aninteresting, concrete, example].[15]

[13] Cummings (2009). The relative merits of risk ratios and odds ratios. Archives of pediatrics & adolescent medicine, 163(5), 438-445.

[14] See e.g., Altman & Sackett. Odds ratios should be avoided when events are common [letter]. BMJ. 1998; 317:1318.

[15] Cook "Advanced statistics: up with odds ratios! A case for odds ratios when outcomes are common." Academic Emergency Medicine 9.12 (2002): 1430-1434.

[16] *Statistical Issues in Drug Development* (3rd Edition), by Stephen Senn (p. 150).

Regression standardization (in a nutshell)

- As an example, the following are steps for one reliable method for covariate adjustment for unconditional treatment effects with binary outcomes that produces a resulting estimator (Steingrimsson et al. 2017; Freedman 2008) termed the "standardized," "plug-in," or "g-computation" estimator:

  (1) Fit a logistic model with maximum likelihood that regresses the outcome on treatment assignments and prespecified baseline covariates. The model should include an intercept term.

  (2) For each subject, regardless of treatment group assignment, compute the model-based prediction of the probability of response under treatment using the subject's specific baseline covariates.

  (3) Estimate the average response under treatment by averaging (across all subjects in the trial) the probabilities estimated in Step 2.

  (4) For each subject, regardless of treatment group assignment, compute the model-based prediction of the probability of response under control using the subject's specific baseline covariates.

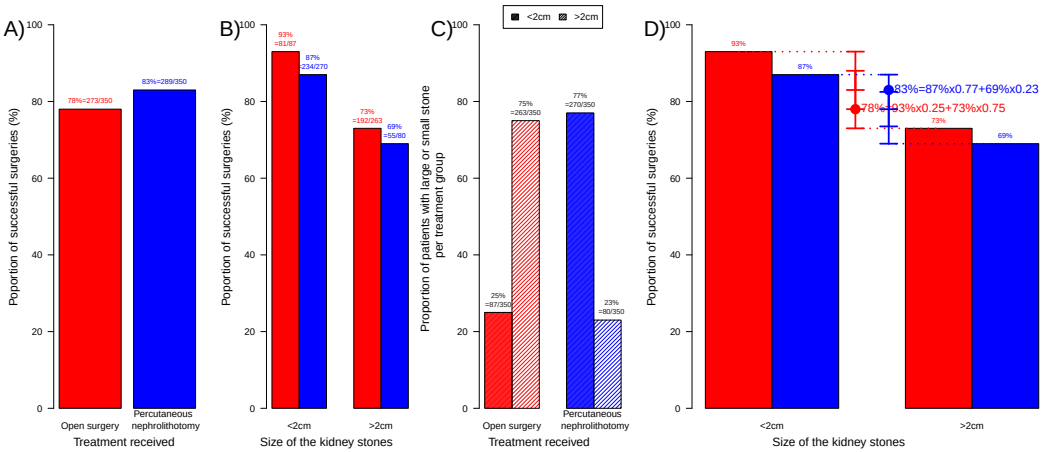
  (5) Estimate the average response under control by averaging (across all subjects in the trial) the probabilities estimated in Step 4.

  (6) The estimates of average responses rates in the two treatment groups from Steps 3 and 5 can be used to estimate an unconditional treatment effect, such as the risk difference, relative risk, or odds ratio.

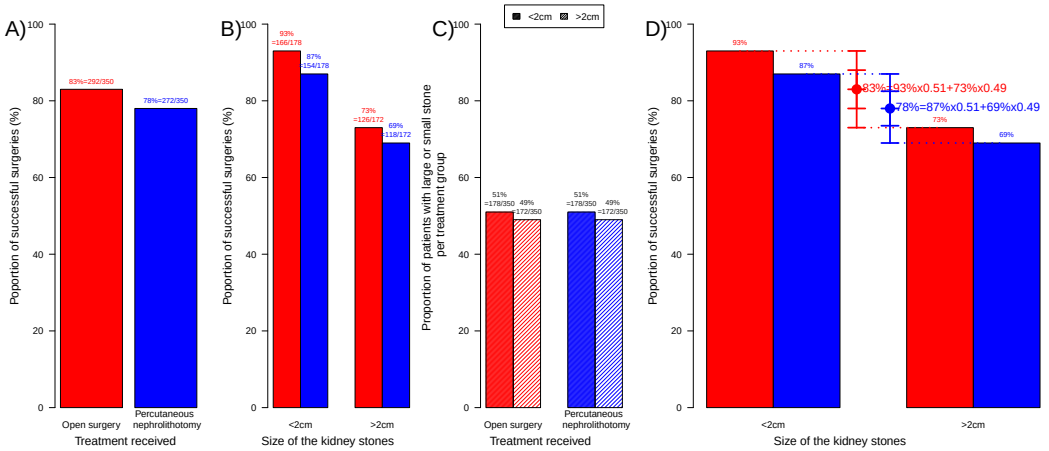
- Recommendation: compute standard errors by bootstrapping.

Source: FDA scientific guidelines: "Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products Guidance for Industry", 2023, accessed 29/10/2025, https://www.fda.gov/media/148910/download

Reminder: Lecture 1, non randomized study.



Similar but randomized (hypothetical data, perfect balance)



- Perfectly balance prognostic covariate: 51% vs 51% small stone in the two arms.
- Idea behind regression standardization: In practice we will almost always observe a bit of random imbalance (e.g., 55% vs 47%). But we can pool the data of the two arms to estimate only one set of weights, instead of one per arm. This reduces the variability of the overall (marginal) risk difference, as any difference directly caused by this imbalance will disappear. This is fine to do so, as the weights are expected to be similar (in average) due to randomization. Another way of thinking about it, is that we can borrow information of data from one arm to estimate the weights of the other arm, because the population weights that we aim to estimate are the same thanks to randomization. Observing different weights in both arms is just due to random variation.

Reminder: logistic regression

Multiple logistic model is a widely used model for the risk of observing a binary outcome $(D = 1)$ given treatment A and baseline covariates $\boldsymbol{L}$:

$$P\{D = 1|\boldsymbol{L}, A\} = \frac{\exp(\boldsymbol{X}^T\boldsymbol{\beta})}{1 + \exp(\boldsymbol{X}^T\boldsymbol{\beta})} ,$$

where $\boldsymbol{X} = \boldsymbol{f}(\boldsymbol{L}, A)$ is a pre-specified vector of variables constructed from $(\boldsymbol{L}, A)$, possibly including interaction terms. Often, $\boldsymbol{X} = (1, A, \boldsymbol{L})$ (i.e., the simplest possible model with no interaction terms, no categorized variables, no splines.)

Reminder: odds ratio and logistic regression

Odds: defined as "risk of event divided by risk of no event".

$$\boxed{\text{odds} = p/(1-p)} ,$$

Odds ratio (OR): defined as the ratio of the odds,

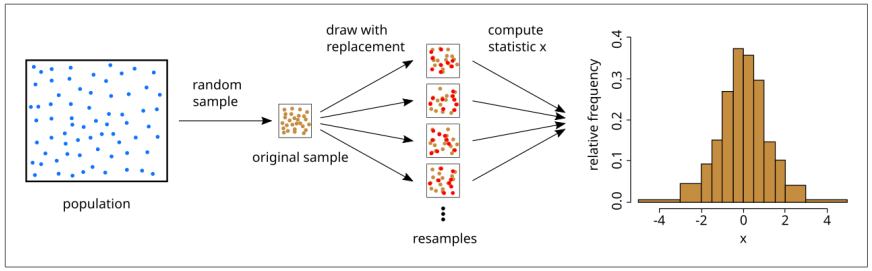
$$\boxed{OR = \frac{\text{odds}_1}{\text{odds}_2} = \frac{p_1/(1-p_1)}{p_2/(1-p_2)}} .$$

With logistic regression (without interactions)

$$e^{\beta_j} = OR_{(X_j = x+1 \text{ vs } X_j = x)}$$

is the odds ratio that compares the risks of two subjects, when both subjects are similar for all covariates adjusted for, except X_j, whith one subject having a value one unit larger than the other subject (i.e. $X_j = x + 1$ vs $X_j = x$).

Digression: bootstrapping (to compute standard errors)



"A sample is drawn from a population. From this sample, resamples are generated by drawing with replacement (orange). Data points that were drawn more than once (which happens for approx. 26.4% of data points) are shown in red and slightly offsetted. From the resamples, the statistic x [e.g., risk difference] *is calculated and, therefore, a histogram can be calculated to estimate the distribution of x [e.g., risk difference]."*

The standard deviation corresponding to that distribution is then computed to estimate the standard error of x [e.g., risk difference].

Source: https://en.wikipedia.org/wiki/Bootstrapping_(statistics), accessed 29/10/2025.
For another explanation, here is a 9:26 mins video: https://youtu.be/Xz0x-8-cgaQ?si=RP-eGVi25ZcxmMAM

Robustness to logistic model misspecification

- The better the fit of the logistic model the larger the power gain.
- *"All models are wrong, but some are useful"* (George Box). The model does not need to be perfect to provide power gains. Sensible models will often do the job almost as well as perfect models.

[16] FDA scientific guidelines: "Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products Guidance for Industry", 2023, accessed 29/10/2025, https://www.fda.gov/media/148910/download

Robustness to logistic model misspecification

- The better the fit of the logistic model the larger the power gain.
- *"All models are wrong, but some are useful"* (George Box). The model does not need to be perfect to provide power gains. Sensible models will often do the job almost as well as perfect models.
- Interestingly, the resulting unconditional treatment effect estimates are unbiased and confidence intervals and p-values are valid (i.e., correct coverage and type-I error) even when the model is misspecified; with large sample sizes. That is, even when the relationship between the outcome and the treatment and baseline variables is it wrongly modeled (e.g., interaction exist but are not included in the model). This is a consequence of randomization. Hence, adjusting for baseline covariates is now widely considered safe:

> *"incorporating prognostic baseline covariates in the design and analysis of clinical trial data can result in a more efficient use of data to demonstrate and quantify the effects of treatment. Moreover, this can be done with minimal impact on bias or the Type I error rate."* [...] *"Covariate-adjusted estimators of unconditional treatment effects that are robust to misspecification of regression models have been proposed for randomized clinical trials with binary outcomes (e.g., Steingrimsson et al. 2017)"*[16]

[16] FDA scientific guidelines: "Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products Guidance for Industry", 2023, accessed 29/10/2025, https://www.fda.gov/media/148910/download

R implementation:

- It is easy![17]
- FDA scientific guidelines[18] cite Steingrimsson et al (2017)[19], which contains easy to use R code in appendix.

Robustness with conditional treatment effect estimation:

- when simply using logistic regression without standardization, we estimate conditional treatment effects, with 95%-CI and p-values.
- Interestingly, a consequence of randomization is that type-I error will also be controlled in that case, even in case of model misspecification; with large sample sizes.[20]
- However, misspecification might introduce bias and 95%-CI might have the wrong coverage in case of model misspecification. This is unfortunate, but "sensible" modelling choices should prevent serious problems.

[17] It is very easy when using the "simplest" possible logistic regression model, without interaction etc, which is relatively often good enough. A few extra lines of code might be needed for more complicated modeling approaches.

[18] FDA scientific guidelines: "Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products Guidance for Industry", 2023, accessed 29/10/2025, https://www.fda.gov/media/148910/download

[19] Steingrimsson et al, *Improving precision by adjusting for prognostic baseline variables in randomized trials with binary outcomes without regression model assumptions*, Contemporary Clinical Trials 54 (2017) 18–24

[20] Rosenblum & van der Laan, 2009, Using Regression Models to Analyze Randomized Trials: Asymptotically Valid Hypothesis Tests Despite Incorrectly Specified Models, Biometrics, 65(3):937-945

A recent example (1/5) (Danish investigator-initiated RCT, completed in 2026)

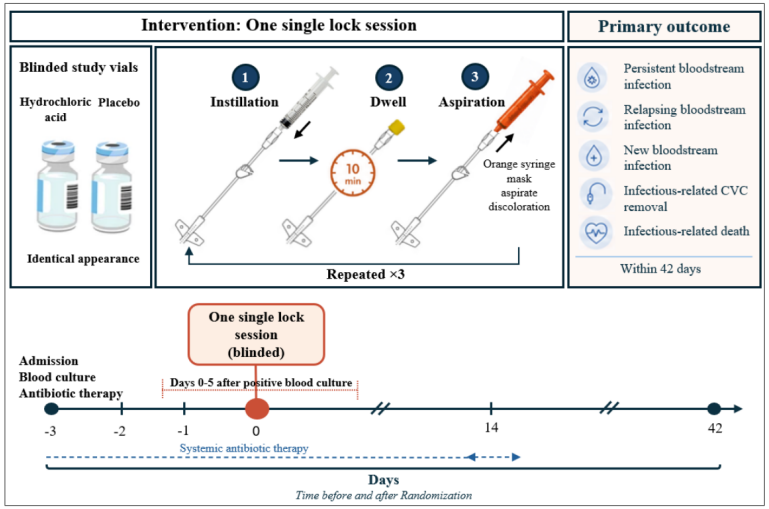
Is hydrochloric acid (HALT) in central venous catheter effective to treat Bloodstream Infections, among children and adults with cancer or hematologic disorders?



(investigator-initiated trial)

- **ClinicalTrials.gov ID:** NCT05376566
- **Link to the statistical analysis plan:** https://cdn.clinicaltrials.gov/large-docs/66/NCT05376566/SAP_000.pdf
- **Acknowledgments:** picture and figure in this slide and the next were provided by Mette Bondo Mønster & Ulrikka Nygaard.

A recent example (2/5) (Danish investigator-initiated RCT, completed in 2026)



A recent example (3/5) (Danish investigator-initiated RCT, completed in 2026)

4.1 Primary objective, endpoint and estimand

The two treatment strategies (interventions) being investigated are the following.

- **Hydrochloric acid lock therapy (HALT)**: Participants receive lock therapy with hydrochloric acid (2.0 molar). The CVAD instillations is carried out by a trained hematologic/oncologic nurse. Hydrochloric acid in a volume corresponding to "dead space" (0.3-1.8 ml) will be instilled in the CVAD and aspirated 10 min later. The instillation is completed three times, as described by Ahmad et al [1].
- **Placebo (normal saline)**: same, but using normal saline instead of Hydrochloric acid.

The procedure requires approximately one hour to be completed. In addition to the study intervention, patients in both arms will receive standard of care, including systemic antibiotic therapy. Blood tests for hemolysis will be collected immediately after the procedure and repeated after 24 hours, together with blood cultures and kidney function tests.

The primary outcome is treatment failure within six weeks after intervention defined as either:

(1) a) persistent infection (persistent positive blood cultures taken 24 hours after HALT/placebo) or b) relapse (a new CLABSI with an identical bacterial isolate)

(2) a new CLABSI with any bacterial isolate

(3) infection-related removal of central access device (CVAD)[1]

[1] Defined as removal of CVAD due to suspected or confirmed CLABSI [20]

(4) infection-related death [2]

Source: statistical analysis plan (SAP) available at: `https://cdn.clinicaltrials.gov/large-docs/66/NCT05376566/SAP_000.pdf`.

A recent example (4/5) (Danish investigator-initiated RCT, completed in 2026)

5.5 Covariate adjustment

The main target of inference is a "marginal" estimand (i.e., a marginal treatment effect) [22, 27], also often termed as "unconditional" [11], similar to that of a simple unadjusted analysis. Although the main analysis will use covariate adjustment, the primary goal of covariate adjustment will be to improve precision in estimating the marginal treatment effect [22, 27].

Covariate adjustment will be performed for the main analysis of the primary outcome and we plan to adjust for study site and type of pathogen, as these variables are expected to be prognostic of the primary outcome; see Sec. 7.1. Note that adults and children were included from different sites, which is the first reason we expect study site to be a relevant prognostic variable. Children and adults have substantially different primary conditions and treatment protocols. Also, CVAD use is not completely similar for adults and children. These differences are suspected to lead to different risks of treatment failure. Second, randomization was stratified by site, which further increases the incentive to adjust for site, in order to follow usual recommendations [11, 18]. Third, one site (Aarhus) used Taurolock, a catheter lock solution with antimicrobial and anticoagulant efficacy, whereas the others did not. No other variables will be adjusted for, as no other factor is suspected to have a strong association with the risk of treatment failure within 6 weeks.

Source: statistical analysis plan (SAP) available at: `https://cdn.clinicaltrials.gov/large-docs/66/NCT05376566/SAP_000.pdf`.

A recent example (5/5) (Danish investigator-initiated RCT, completed in 2026)

We will use logistic regression together with standardization to estimate the six weeks risks in each arm with two-sided 95%-CIs, their difference with two-sided 95%-CI and the corresponding p-value. This approach corresponds to using the standardized estimator advocated in Steingrimsson et al. [27]. Standard errors will be computed by non-parametric bootstrapping. This approach is promoted and outlined in a recent FDA guidance document [11]. Specifically, we will adjust for two categorical variables: study site (three levels) and type of pathogen (two levels), as already discussed in Sec. 5.5. The study site variable adjusted for will include only three levels, not four, as we will merge the two levels corresponding to the pediatric department of Odense and Copenhagen University Hospitals. The rationale for merging is that the department of this two sites work in very close collaboration and because the sample size from Odense is not large. The type of pathogen (pathogenic or less pathogenic) is defined as described in Section 10.2.3. It is expected to be prognostic of treatment failure, for instance via a difference in virulence or response to HALT. These variables will be included in the multiple logistic model in addition to the binary variable that indicates the randomized assignment to HALT or placebo, according to the ITT principle. Computation of the standardized estimator and its robust standard error via bootstrap will be performed as described in appendix A.5 in Steingrimsson et al. [27] (with 1000 bootstrap samples).

Source: statistical analysis plan (SAP) available at: `https://cdn.clinicaltrials.gov/large-docs/66/NCT05376566/SAP_000.pdf`.

Practice with R!

Practice with the "Exercise.2.1" from the webpage.
It is about thrombolytic strategies for acute myocardial infarction.